

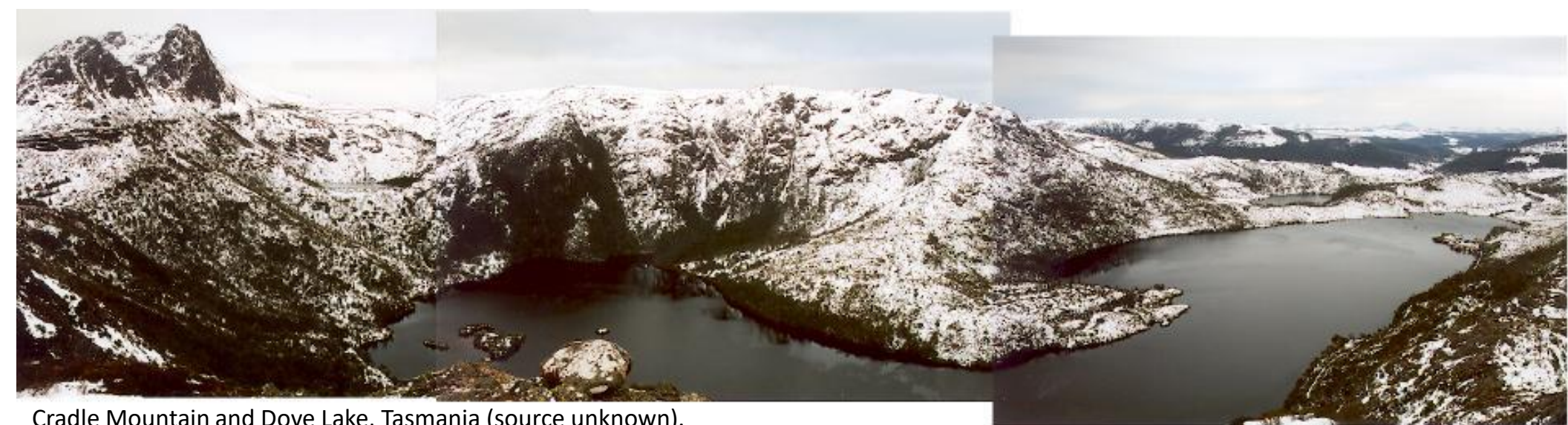


Mt Wellington, Hobart (source unknown).



Mt Buller, Victoria (source unknown).

Snow Forecasting



Cradle Mountain and Dove Lake, Tasmania (source unknown).

Learning Outcomes

- Explain how to distinguish snow, freezing rain and ice pellet environments on the basis of a vertical sounding and understand the physical reasons for these distinctions.
- Describe the hazards represented by these conditions.
- Explain why snow can sometimes still fall in environments where the surface temperature is slightly above freezing.
- Describe the basis of the Partial Thickness Technique.
- Describe the basis of the Dave Williams Snow Probabilities used in Visual Weather and GFE.

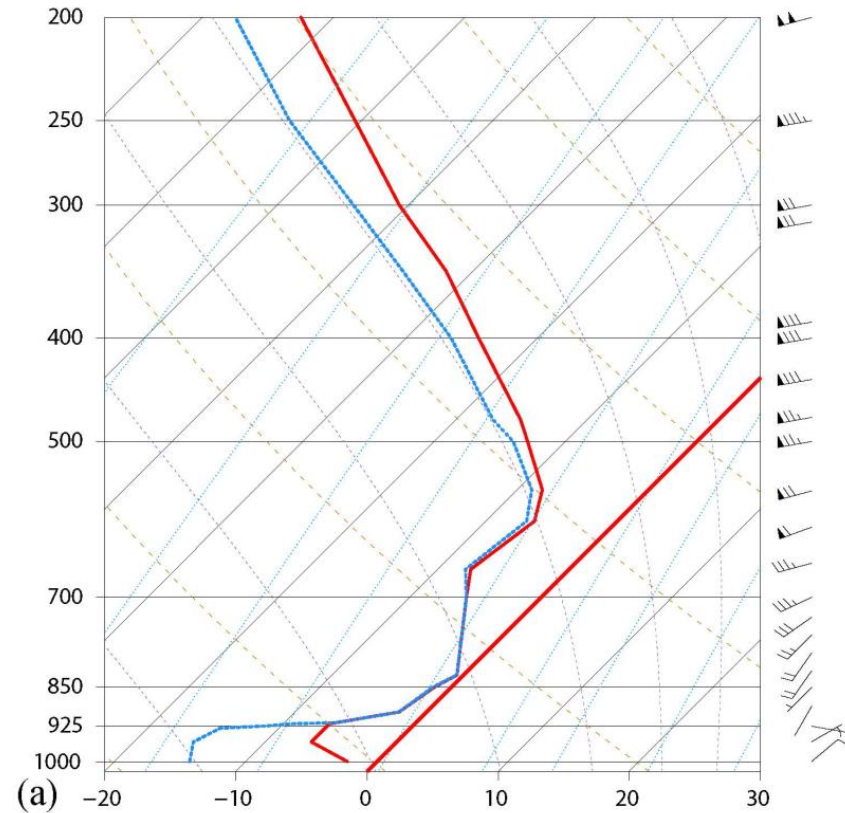
Snow

AMS Definition:

Precipitation composed of white or translucent ice crystals, chiefly in complex branch hexagonal form and often agglomerated into snowflakes.

In practice:

Always forecast snow if it is guaranteed that ice crystals have not been subject to any melting before reaching the ground... (and also in some other instances.)



Lackmann Figure 9.3(a) p222. Profile from Greensboro, North Carolina 1800 UTC 4th December 2002.

Impacts of Snow (Lackmann p220)

- "For seasoned winter storm veterans living in cities characterised by appreciable annual snowfall, efficient snow-removal strategies and experienced drivers lessen the impact of winter storms to some extent.
- However, for major metropolitan areas and in regions that are climatologically less accustomed to winter weather, snowstorms are often associated with significant economic and societal disruptions:
 - Travel delays and cancellations
 - Damage to utilities and infrastructure
 - School closures
 - Reduced access to emergency services
 - Event rescheduling and retail losses

Wake Co. NC schools closed for 2 weeks



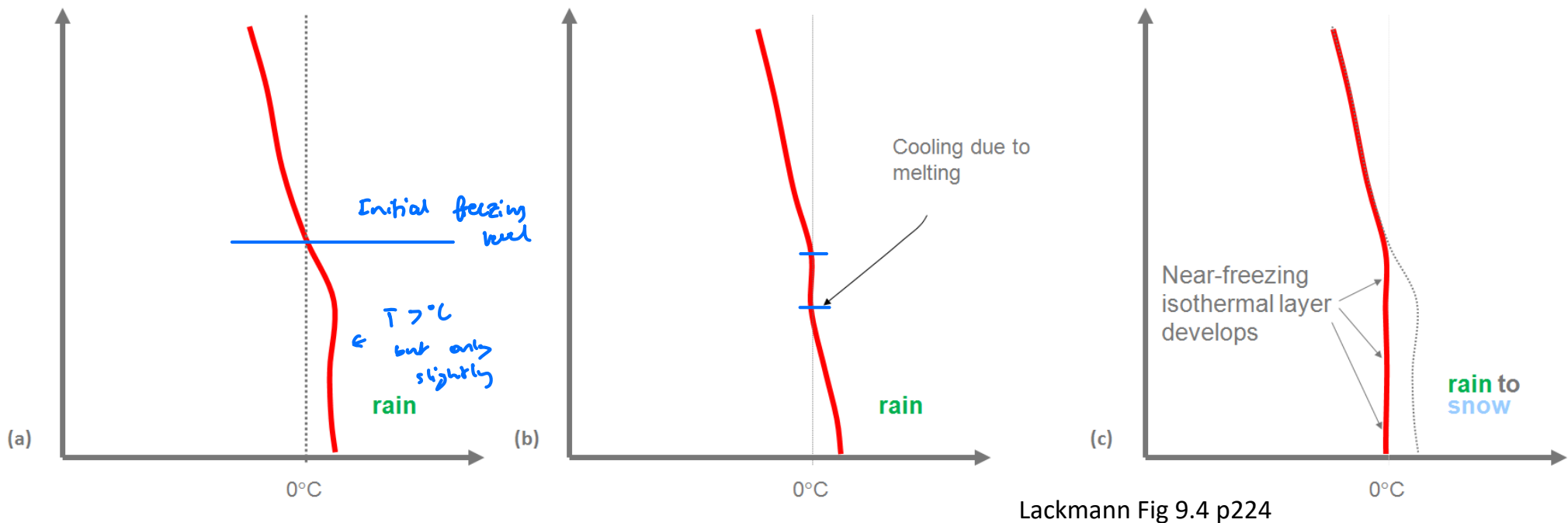
Lackmann Fig. 9.1(c) p220 Hillsboro Street, 25 January 2000 (photo by Neil Jacobs)

1mm Rain = How much Snow?

- Standard rule of thumb: 1mm of rain is equivalent to 1cm of snow.
- Measurements of freshly fallen snow indicate that the snow to rain ratio can vary from 3:1 to 100:1.
- Forecasting snowfall is a two-step problem:
 - Amount of liquid water to fall (Quantitative Precipitation Forecast “QPF problem”)
 - Conversion of liquid water to snow (“snow density problem”).
- Snow density can depend upon a number of factors:
 - In-cloud: Crystal habit and size, degree of riming and aggregation.
 - Sub-cloud: Melting and sublimation.
 - Surface processes: Compaction and melting/sublimation in snowpack.
- An unresolved problem, neural network approaches currently being tried:
[Roebber et al, Improving Snowfall Forecasting by Diagnosing Snow Density, Weather and Forecasting Volume 18 p264, AMS.](#)

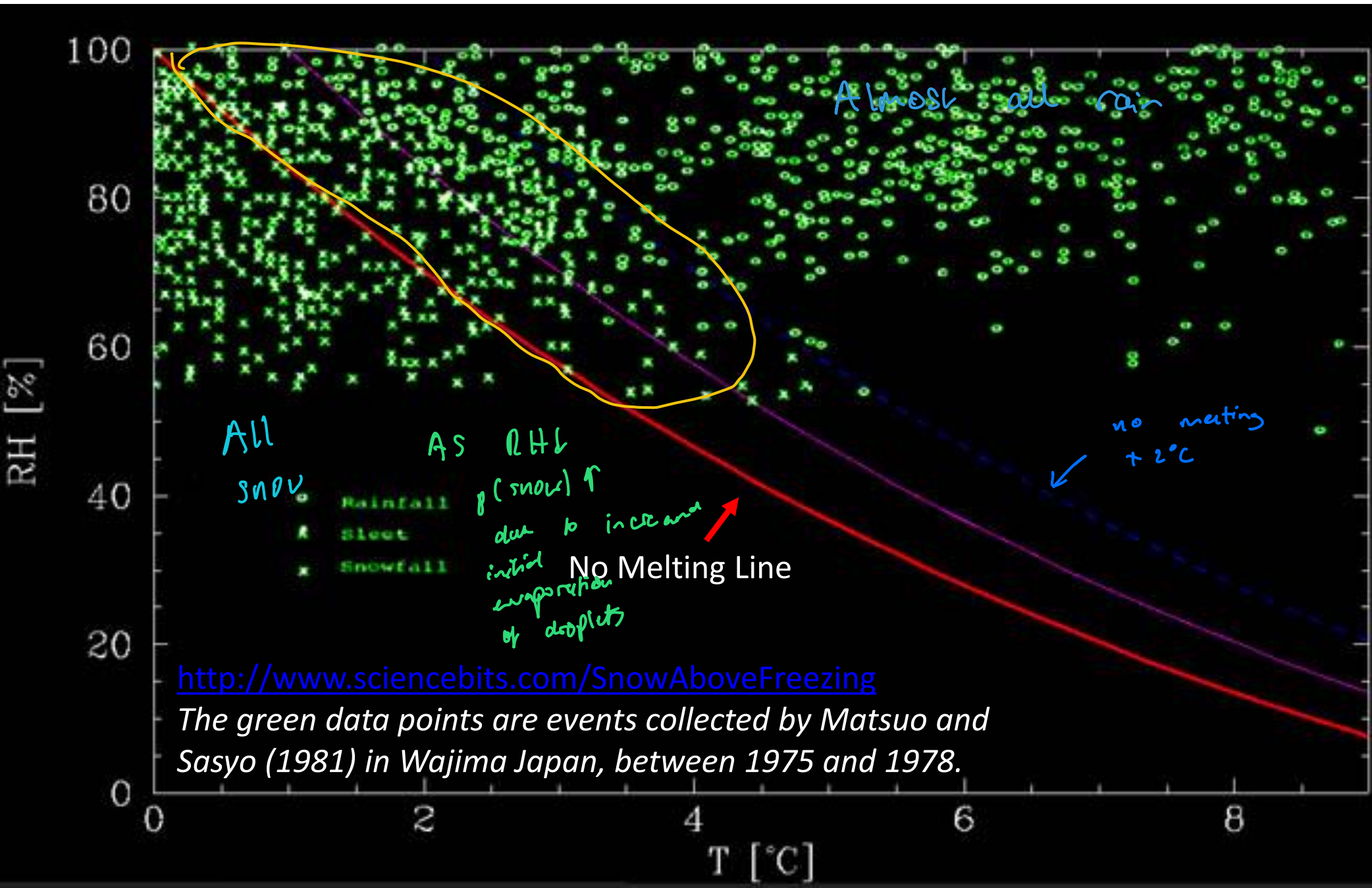
Warm Snowstorms (SFC T > 0)

- If the dominant diabatic process in the near-surface layer is cooling due to melting, a temperature profile that is initially > 0°C can drop to near-freezing. Precipitation that begins as rain can tend to snow as a result.



- Alternatively, if the surface temperature is > 0°C but the near-surface air is very dry, precipitation falling into this layer will evaporate, causing enough cooling to keep snowflakes from melting.

Precip. Type as Function of T,RH



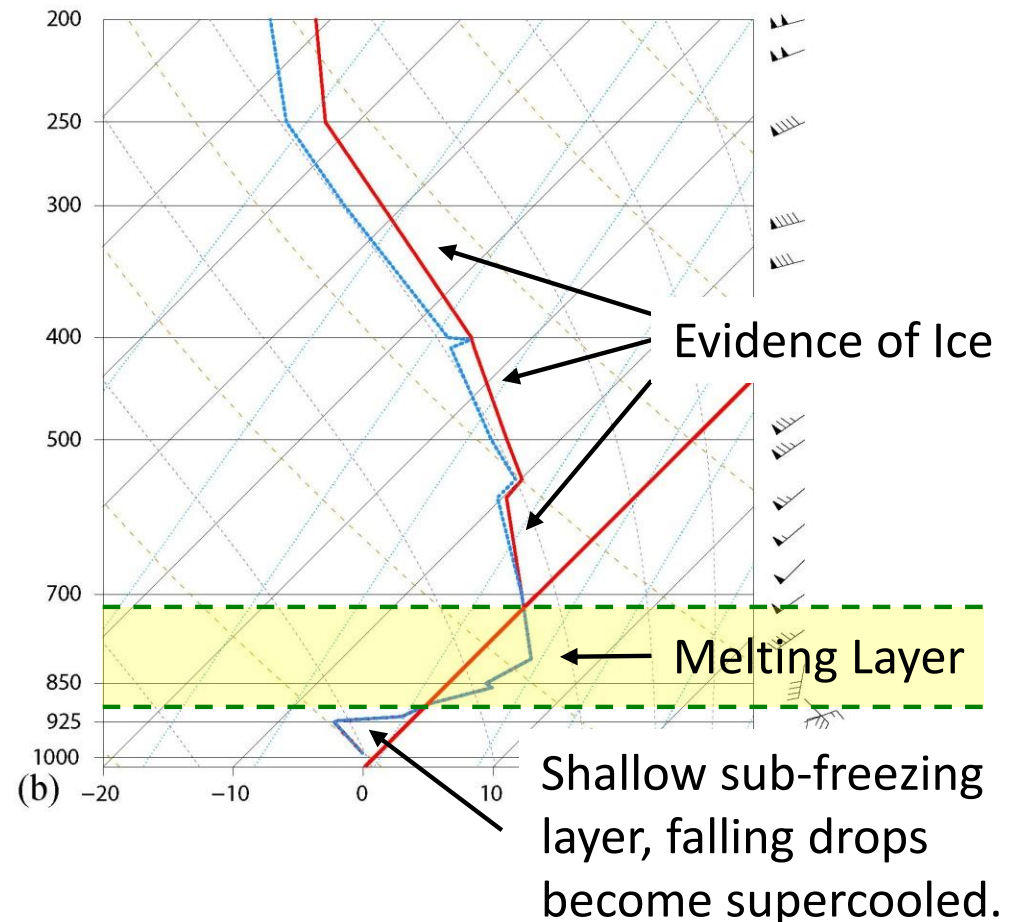
Freezing Rain

AMS Definition:

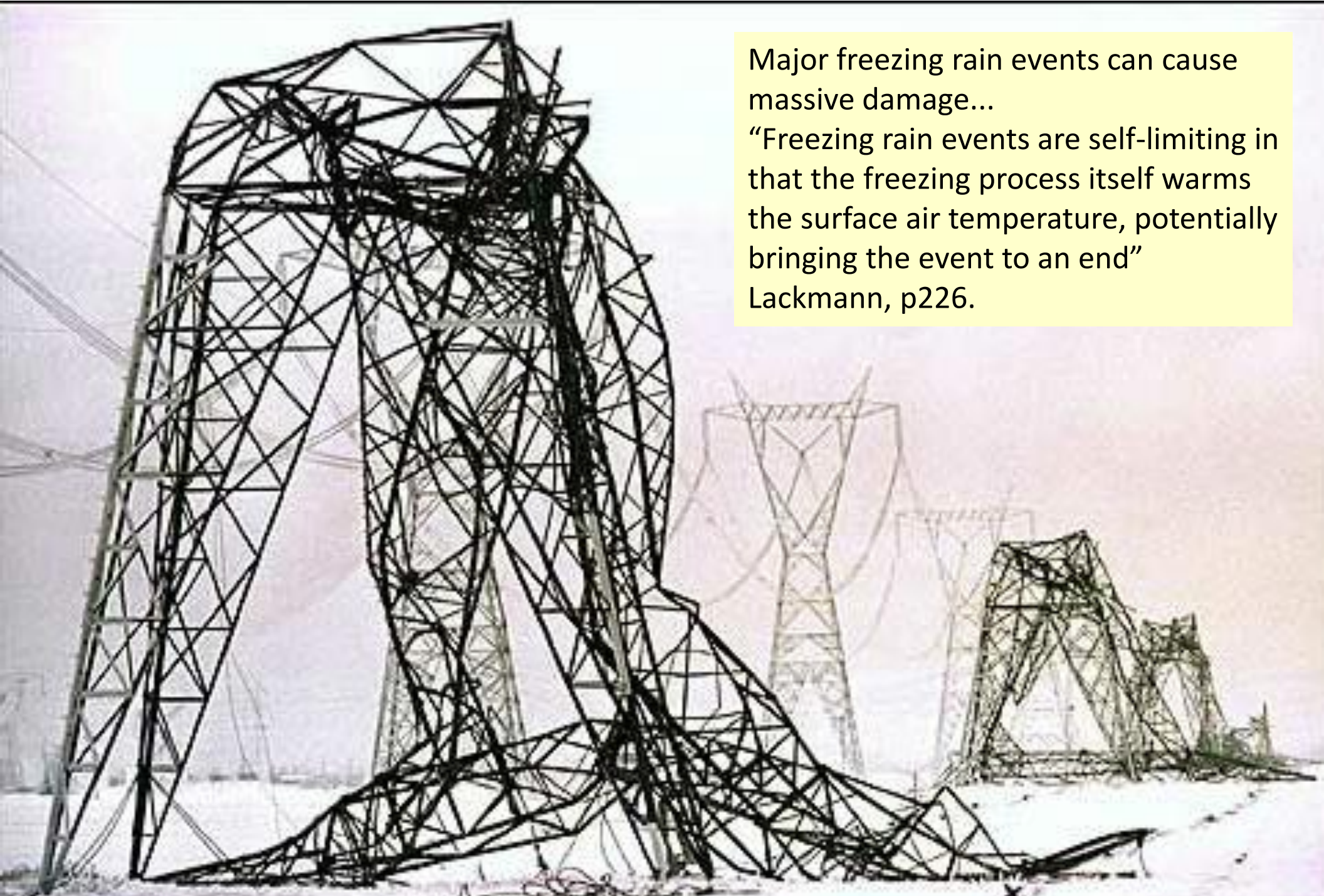
Rain that falls in liquid form but **(becomes supercooled and)** freezes upon impact to form a coating of glaze upon the ground and on exposed objects.

In practice:

Requires an elevated warm layer ($T > 0^{\circ}\text{C}$) of sufficient depth to melt falling ice and a shallow sub-freezing layer beneath for the droplets to become supercooled but not frozen.



Lackmann Figure 9.3(b) p222. Profile from Greensboro, North Carolina 0600 UTC 5th December 2002.



Major freezing rain events can cause massive damage...

“Freezing rain events are self-limiting in that the freezing process itself warms the surface air temperature, potentially bringing the event to an end”

Lackmann, p226.

Lackmann Figure 9.1 (a) p220. Collapse of transmission line towers near St. Bruno, Quebec, during the January 1998 ice storm that left more than 1.4 million people without power.

Ice Pellets

AMS Definition:

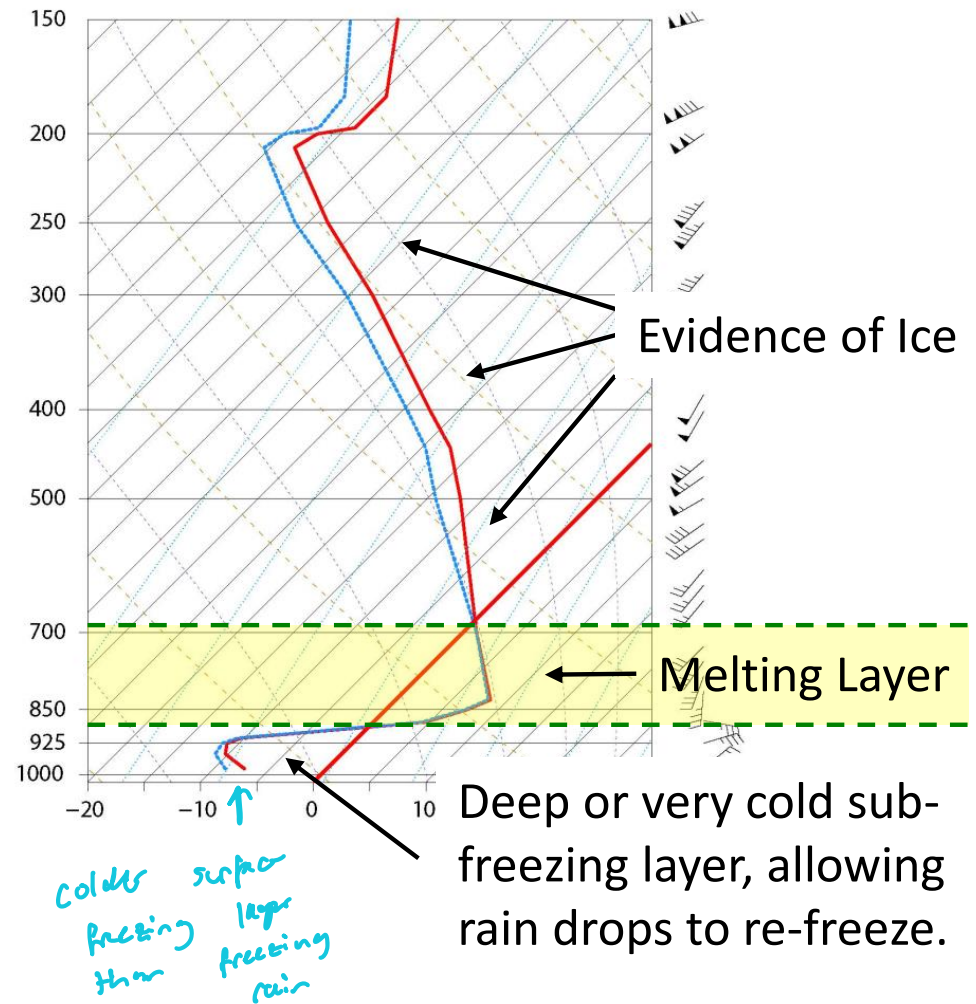
A type of precipitation consisting of transparent or translucent pellets of ice, less than 5 mm in diameter.

Ice pellets usually bounce when hitting hard ground and include:

- 1) Sleet or grains of ice formed by freezing rain drops.
- 2) Small hail.

In practice:

Requires a deeper or colder sub-freezing layer than for freezing rain.



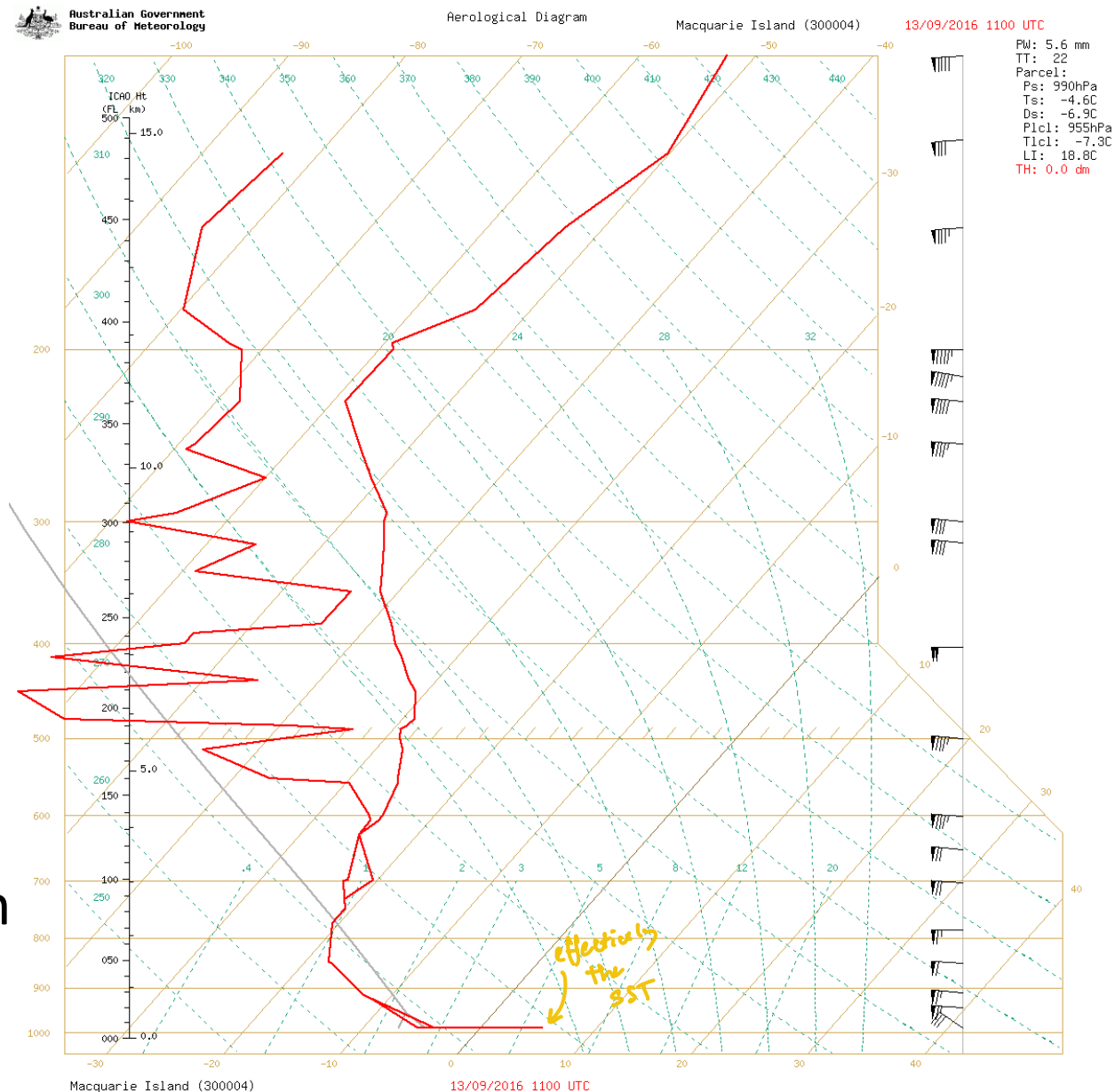
Lackmann Figure 9.9 p228. Profile from Greensboro, North Carolina 0000 UTC 17th February 2003.

Although accumulations of several cm can occur, "the absence of a glaze renders these events less damaging than an event characterised by the same liquid-equivalent amount of freezing rain".
Lackmann, p225.



Snow in Australia

- Snow at sea-level is exceedingly rare in mainland Australia.
- Normally only a concern for alpine areas ($>1200\text{m}$) in winter.
- Snow at sea-level can occasionally occur in Tasmania.
- A routine consideration for Macquarie Island and Antarctica.



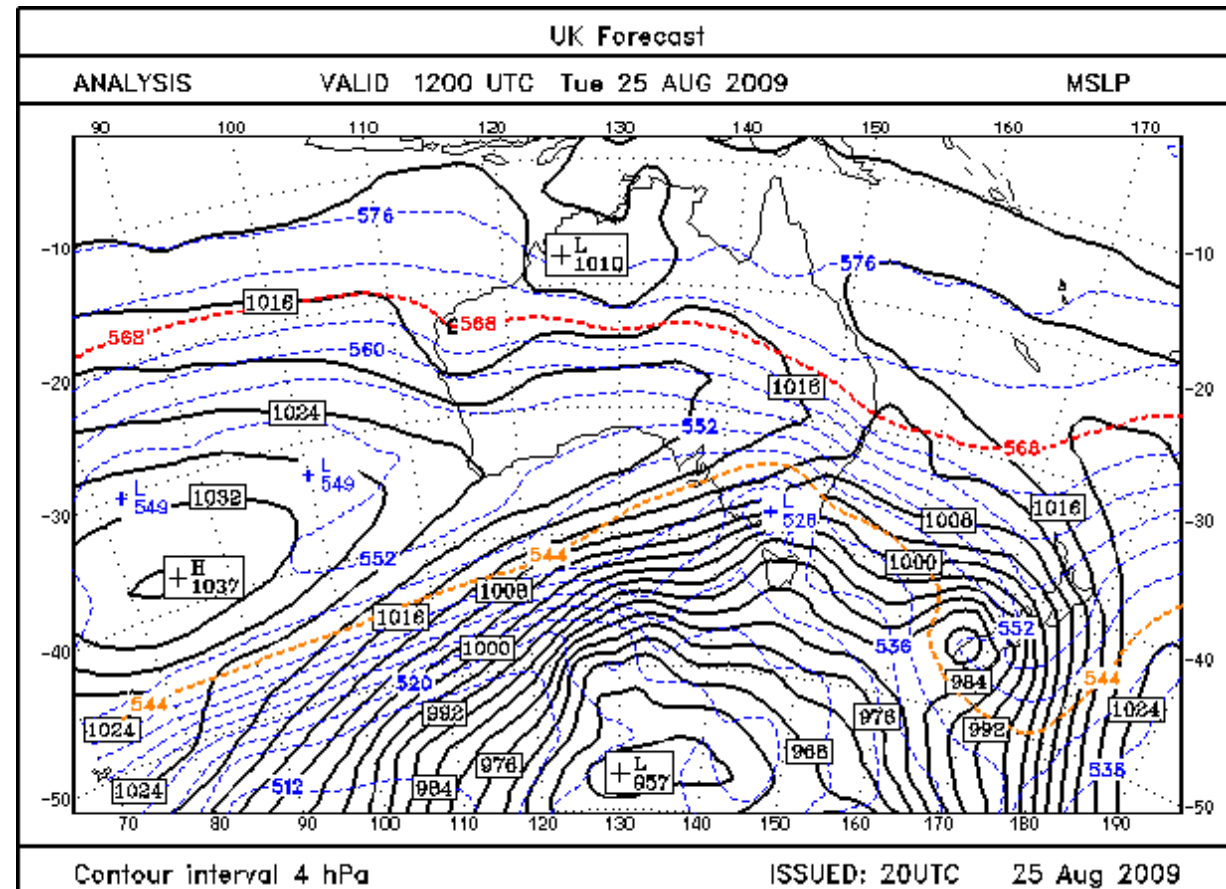
There are 4 main snow producing situations for southeast Australia

Upper cold pool

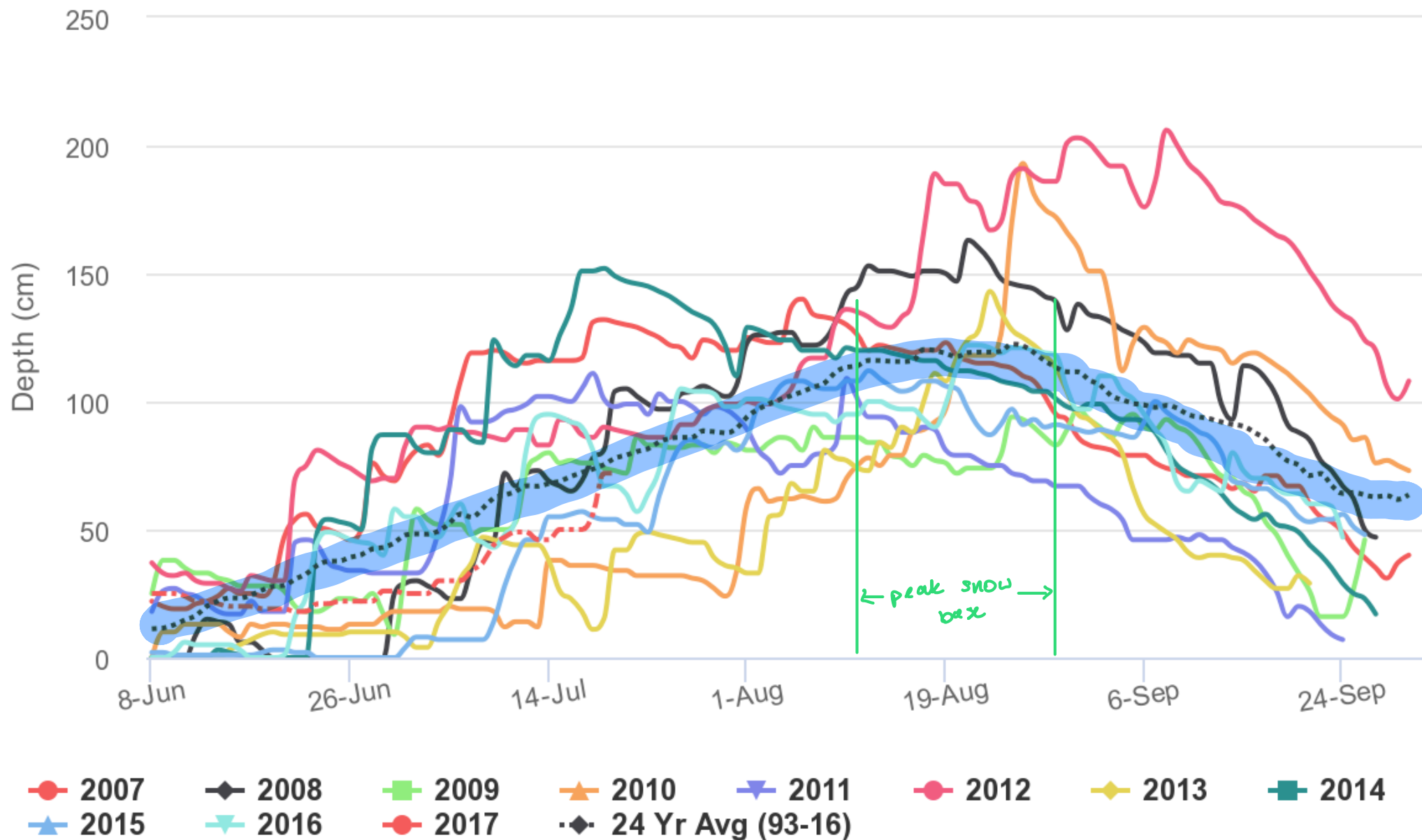
Pre-frontal NW stream

W stream

Cold S outbreak



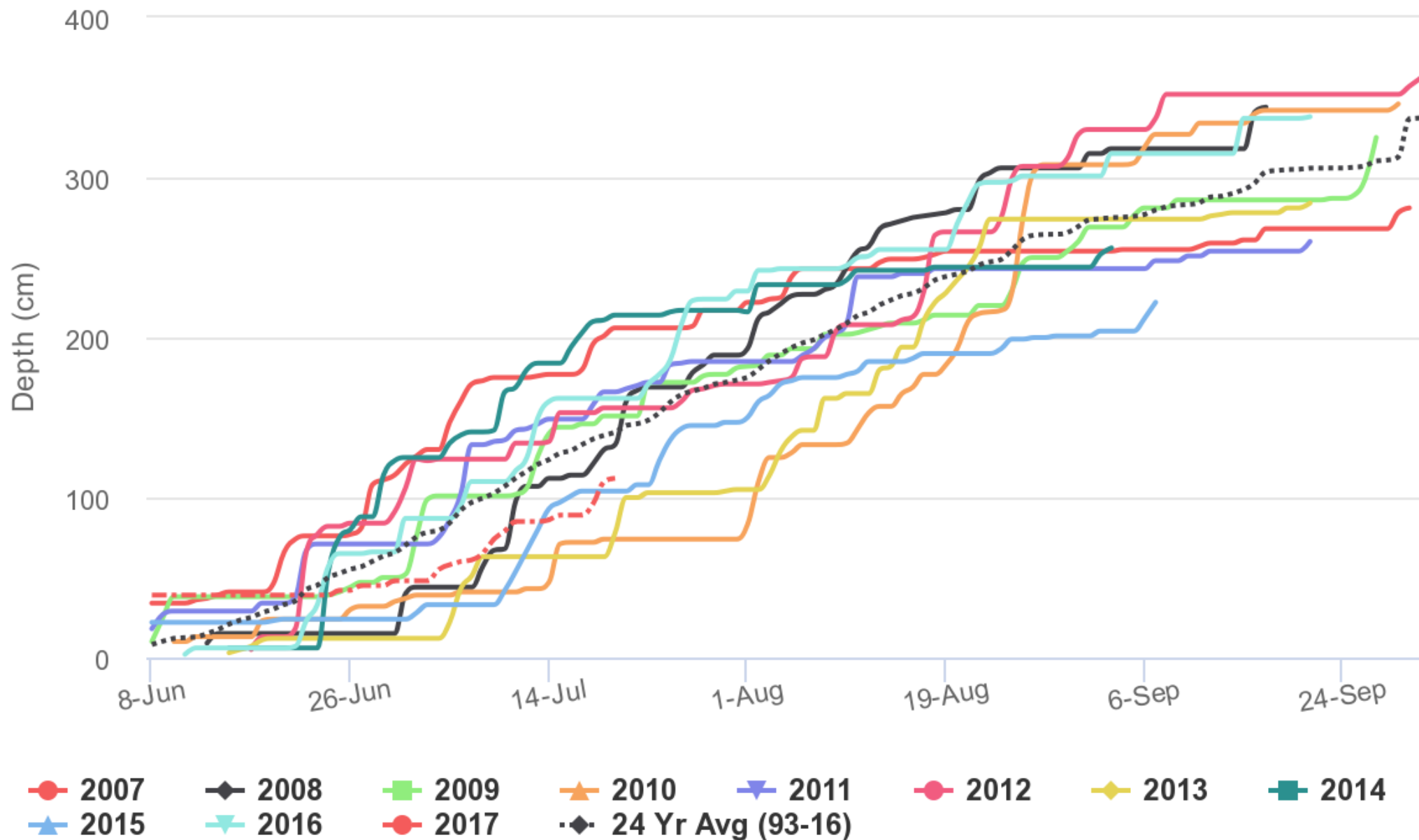
Hotham Natural Snow Base - Historic View



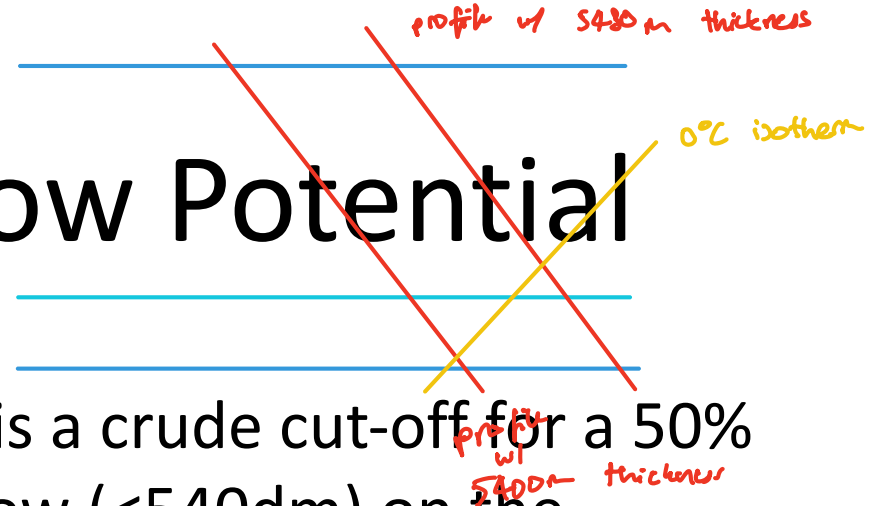
Highcharts

<http://www.mthotham.com.au/all-about-hotham/snow-weather/snow-charts/>

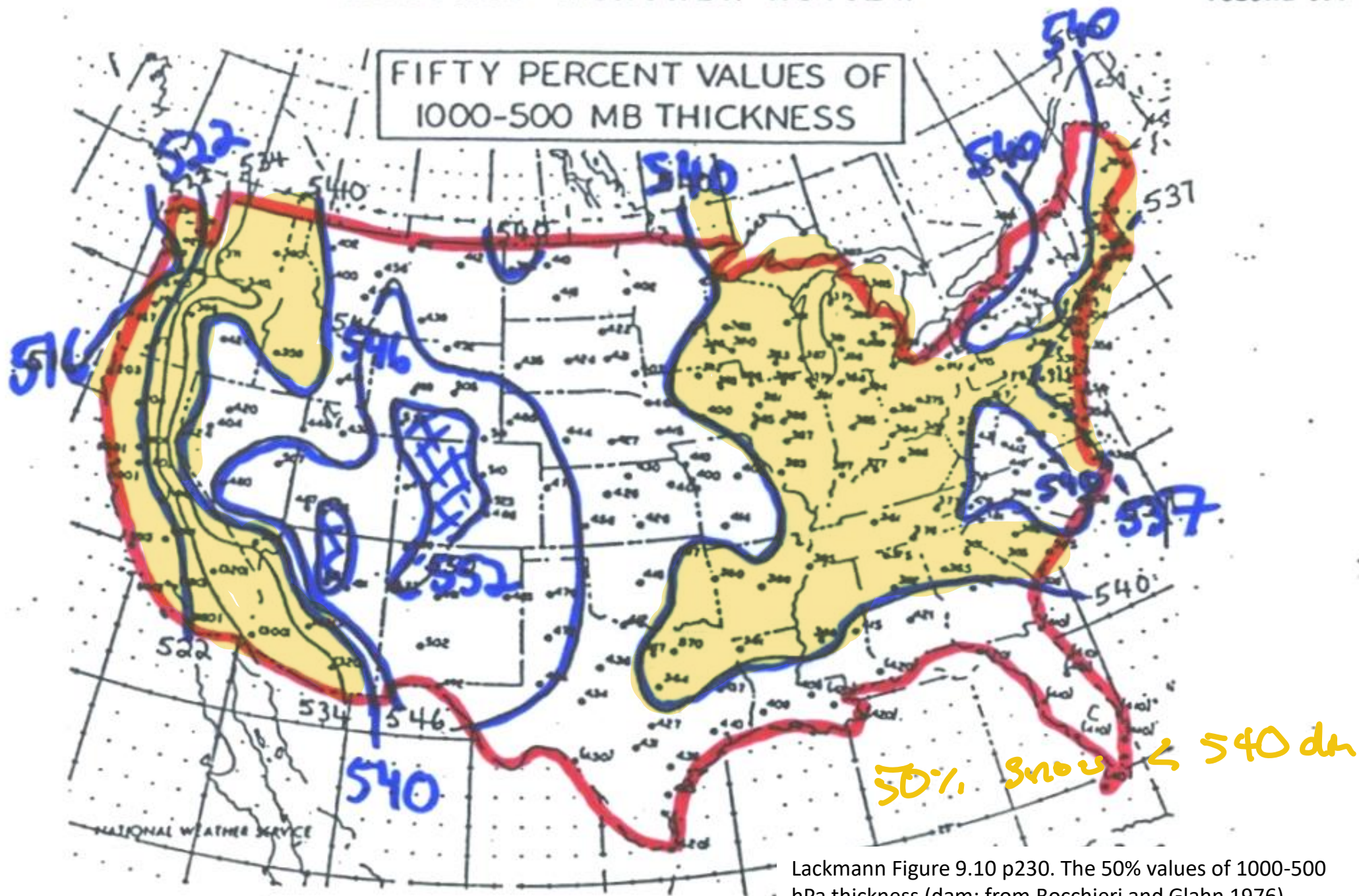
Hotham Accumulated Snow Falls - Historic View



Thickness and Snow Potential



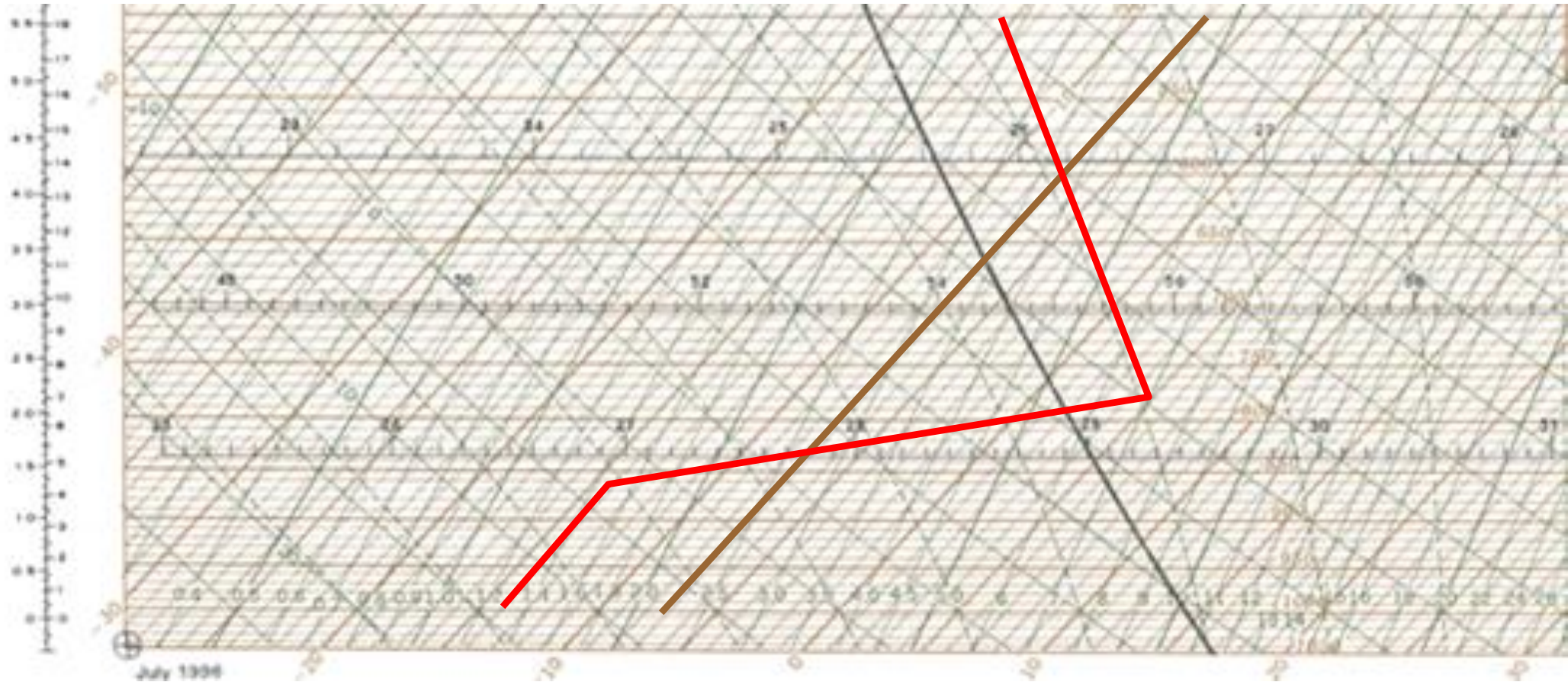
- 1000-500 hPa thickness of 540dm is a crude cut-off for a 50% chance of rain ($>540\text{dm}$) versus snow ($<540\text{dm}$) on the ground.
- In elevated terrain, the rain/snow threshold occurs at higher thickness, $\Delta z > 540\text{dm}$.
- In areas where the surface temperature is warmer, including maritime regions, the rain/snow cut-off is lower than 540dm.
- However, a vast number of temperature profiles are possible for a thickness of 540dm, many are not conducive to snow.
- In mainland Australia, higher surface temperatures mean that snow at sea level requires thickness closer to 524dm (rare!).



Lackmann Figure 9.10 p230. The 50% values of 1000-500 hPa thickness (dam; from Bocchieri and Glahn 1976).

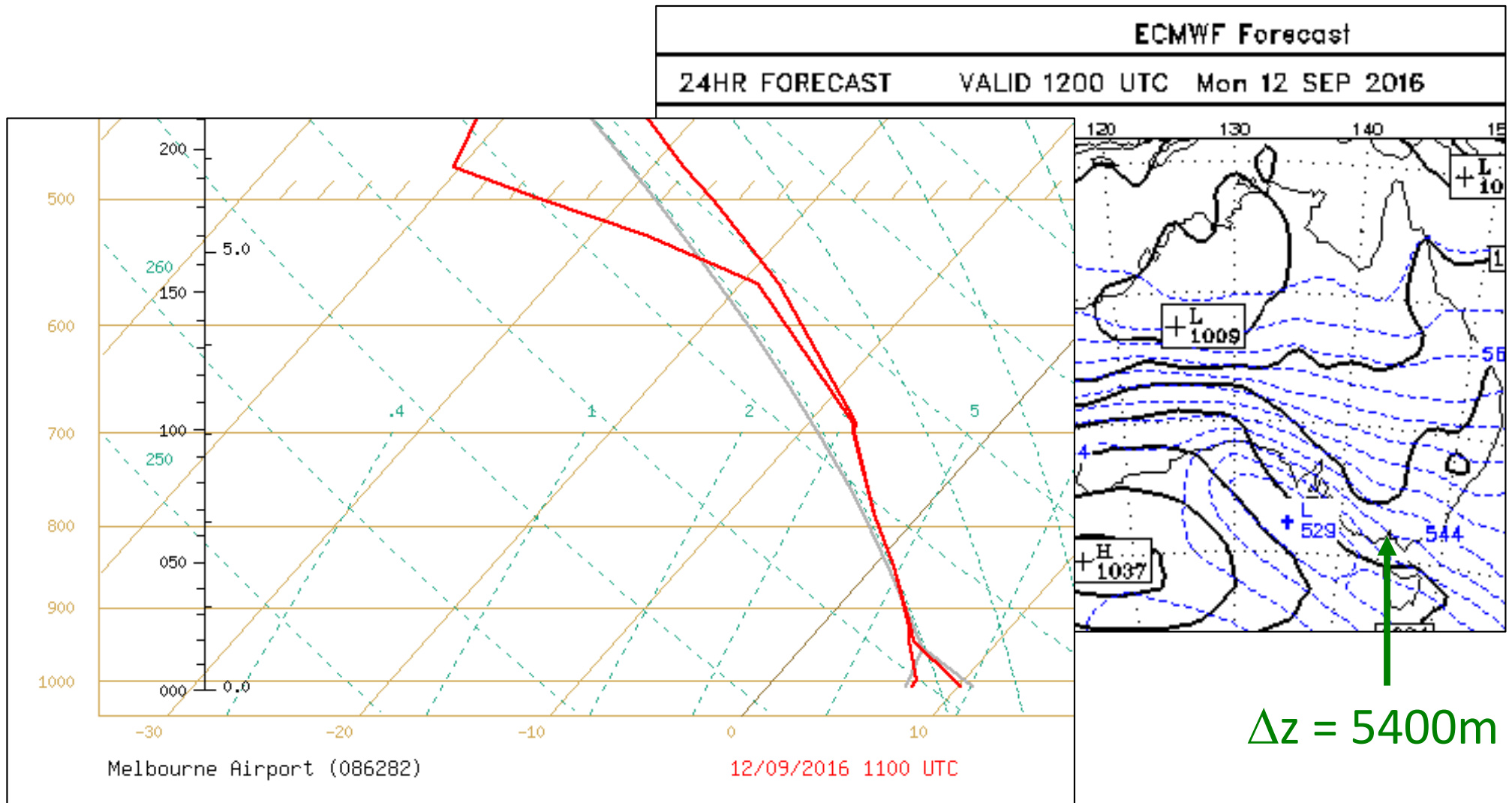
FIG. 1. Fifty-percent values of 1000-500 mb thickness in meters with the leading 5 omitted. Estimates for stations for which 50% values could not be determined statistically are in parentheses.

5400 Thickness with high stability



Winter weather soundings in eastern USA show similarities with this idealised profile. The surface temperature low enough for freezing precipitation.

5400 Thickness with low stability



Winter weather soundings in southern Australia (and maritime regions) have more in common with this profile. The surface temperature is too high for freezing precipitation.

Partial Thickness Technique

Thicknesses generally corresponding to the location of a rain/snow boundary in a non-marine environment in North America

Layer (hPa)	Thickness (m)
1000-500	5400
1000-700	2840 (+ or - 20)
1000-850	1290
850-700	1540
850-500	4100
700-500	2560

Snow becomes a consideration about alpine peaks in southeast Australia if 1000-500 hPa thickness < 5480 m.

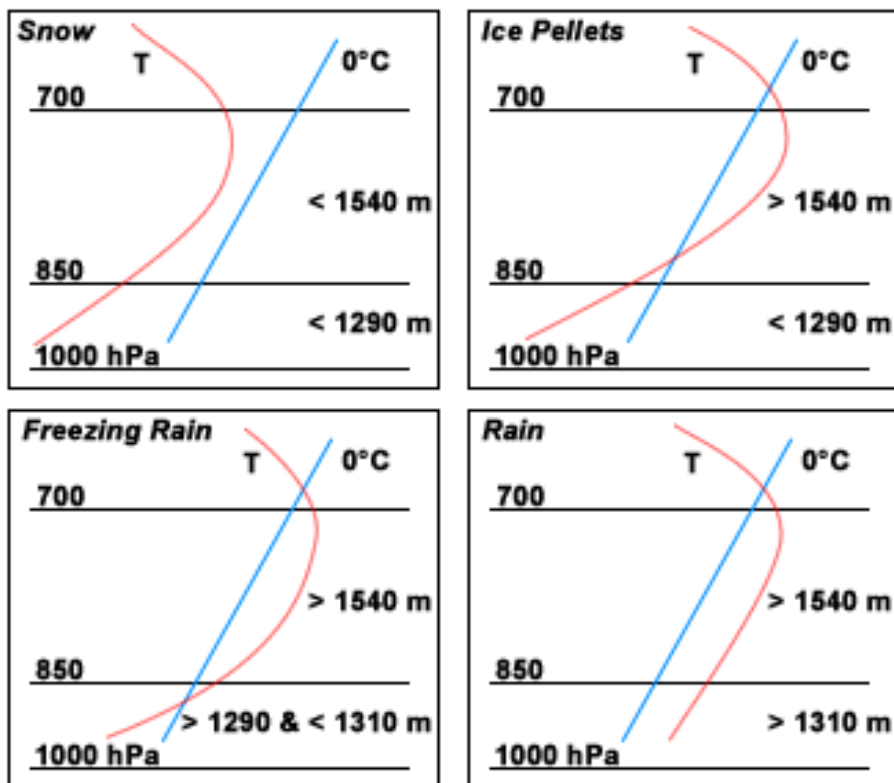
Thicknesses generally corresponding to the location of a rain/freezing rain or ice pellet boundary

Layer (hPa)	Thickness (m)
U.S.A.	
1000-850	< 1295
850-700	> 1555
Ontario	
1000-850	< 1314 +/- 6
850-700	> 1541 +/- 7
Maritimes	
1000-850	< ~ 1300
850-700	> 1540

- For high elevation stations the surface pressure is < 1000 hPa.
- The 1000-500 hPa Thickness values are usually obtained by extrapolation assuming a standard atmospheric lapse rate.
- This lower “warmer portion” does not actually exist. Thus they can receive snow for 1000-500 hPa thickness > 5400 m.

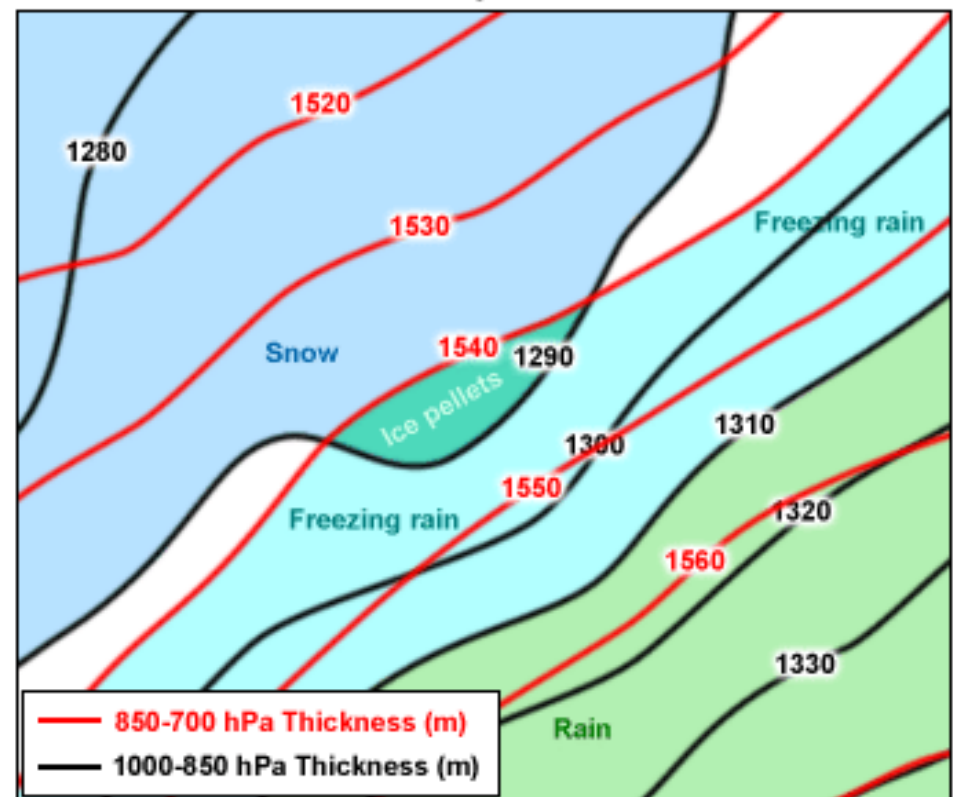
Partial Thickness Technique

Typical 700 - 850 and 850 - 1000 hPa Thickness (m)
Associated with Precipitation Types



Environment Canada

Application of Partial Thickness Method
in a Mixed Precipitation Scenario

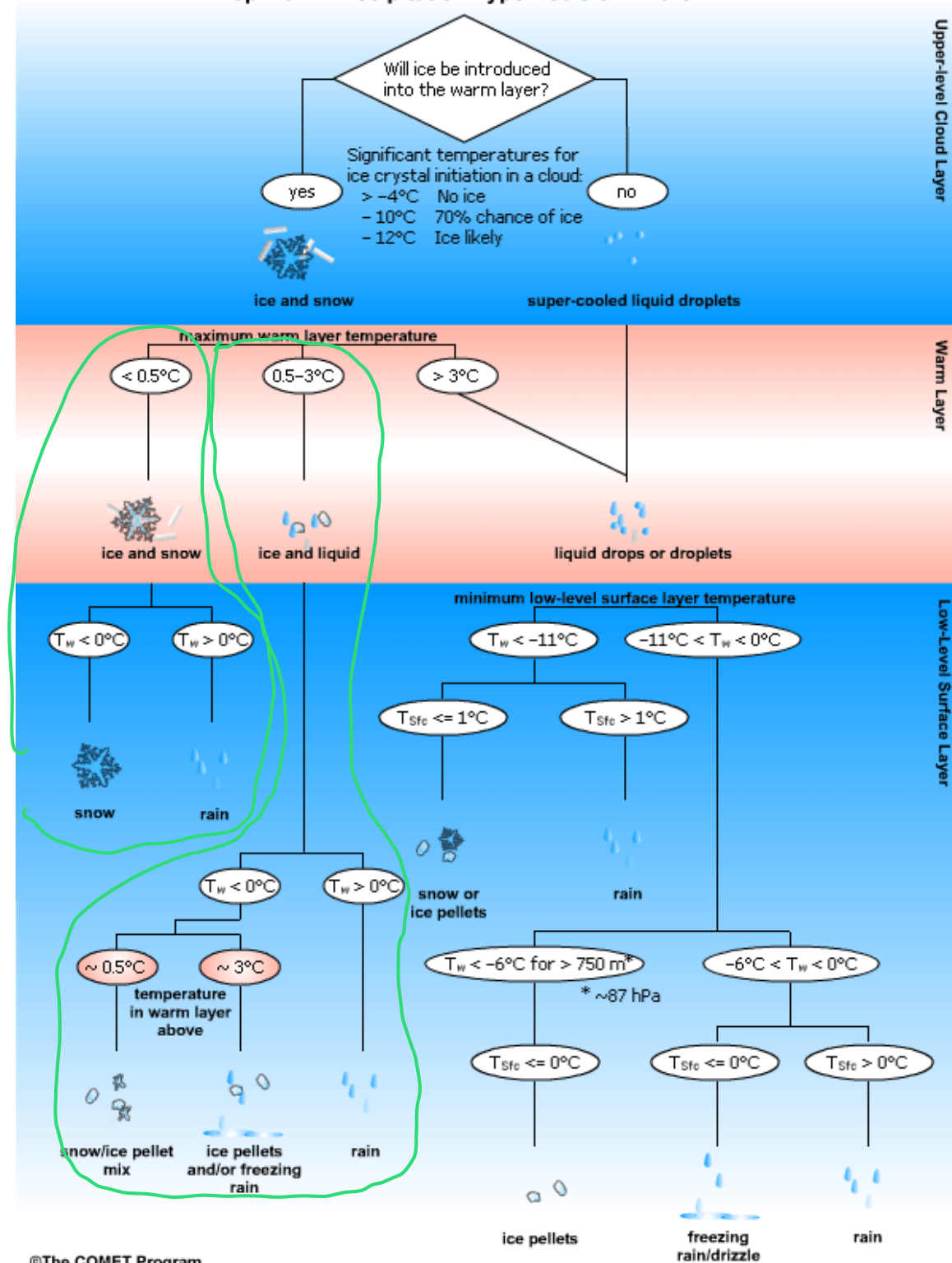


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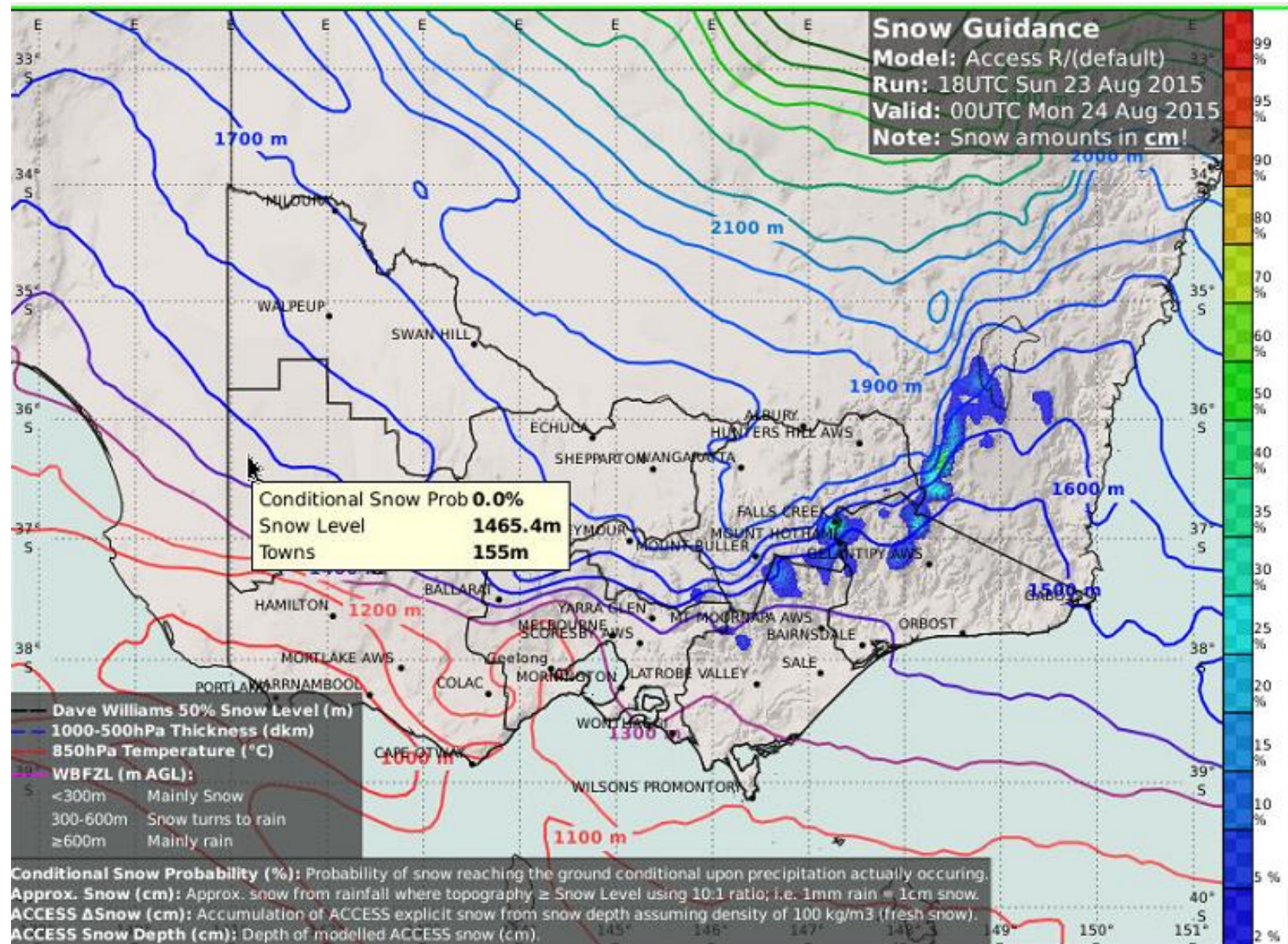
Top Down Method

- Apply on a sounding-by-sounding basis to evaluate the type of freezing precipitation.
- Criteria for each of:
 - Upper level cold layer
 - Elevated warm layer
 - Near surface layer
- Easy to follow rules-of-thumb for the precipitation type based on temperature and depth of near surface layer.

Top-Down Precipitation Type Decision Chart



Dave Williams Snow Probabilities



Dave Williams Snow Probabilities

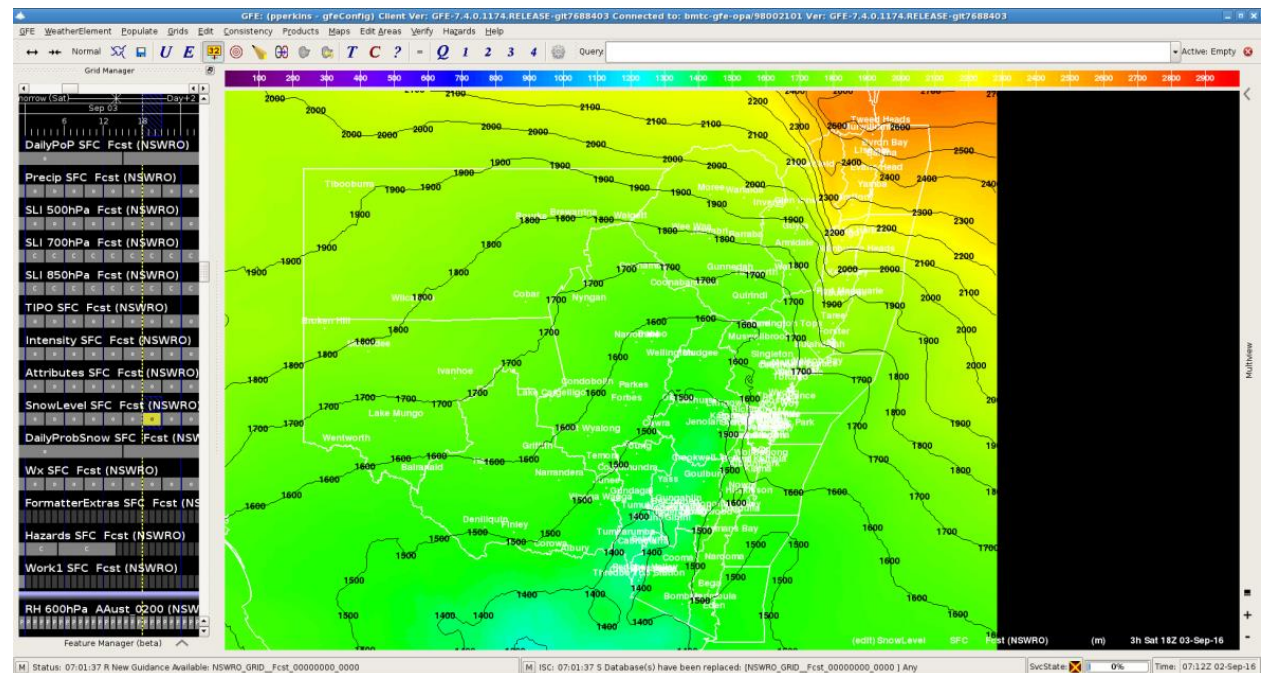
For Station Locations:

Given that is precipitating what is the probability that it will be snow?

Station	Height(m)	0	+	24	+	48	+	72	+	96	+	120	+	144	+	168
CANBERRA AP	578	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TARALGA PO	845	0	0	0	0	0	0	0	0	1	7	0	0	0	0	0
YASS	520	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
COOMA AP AWS	930	0	0	0	0	0	0	0	28	3	2	0	0	0	0	0
TIDBINBILLA	743	0	0	0	0	0	0	0	4	1	0	0	0	0	0	0
BOMBALA AWS	761	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0
GOULBURN AP AWS	640	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
TUGGERANONG	587	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
MT GININI AWS	1760	1	0	0	0	53	75	99	97	85	0	0	0	0	0	0
CHARLOTTE PASS	1755	0	0	0	0	69	69	99	97	92	0	0	0	0	0	0
CRACKENBACK	1957	3	1	2	91	91	99	99	98	0	0	0	0	0	0	0
THREDBO VILLAGE	1380	0	0	0	0	11	11	96	71	43	0	0	0	0	0	0
PERISHER VALLEY	1735	0	0	0	0	64	65	99	97	91	0	0	0	0	0	0
TUMBARUMBA PO	645	0	0	0	0	0	0	6	1	0	0	0	0	0	0	0
CABRAMURRA	1482	0	0	0	0	16	27	97	83	49	0	0	0	0	0	0
BARRABA PO	500	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WOOLBROOK	910	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0
GLEN INNES PO	1062	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TENTERFIELD	870	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GUYRA HOSPITAL	1332	0	0	0	0	0	0	0	0	21	0	0	0	0	0	0
ARMIDALE AP AWS	1080	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0
INVERELL	582	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TABULAM	555	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NULLO MOUNTAIN	1130	0	0	0	0	0	0	0	1	69	0	0	0	0	0	0
KATOOMBA	1030	0	0	0	0	0	0	0	1	45	0	0	0	0	0	0
LITHGOW	950	0	0	0	0	0	0	0	0	33	0	0	0	0	0	0
ORANGE AG INST	922	0	0	0	0	0	0	0	2	18	0	0	0	0	0	0
BATHURST AP AWS	745	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0
MT BOYCE AWS	1080	0	0	0	0	0	0	0	1	56	0	0	0	0	0	0
OBERON	1190	0	0	0	0	0	0	0	8	70	1	0	0	0	0	0
ORANGE AP AWS	947	0	0	0	0	0	0	0	2	22	0	0	0	0	0	0
COONABARABRAN	505	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BOWRAL	690	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0
NERRIGA	630	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
BRAIDWOOD	665	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
STRATHBOGIE	520	0	0	0	0	0	0	0	24	1	0	0	0	0	0	0
HUNTERS HILL	981	0	0	0	0	0	0	1	71	22	2	0	0	0	0	0
MT BULLER	1707	0	0	0	1	79	78	99	99	92	0	0	0	0	0	0
OMEQ	685	0	0	0	0	0	0	0	33	3	0	0	0	0	0	0
FALLS CREEK	1767	0	0	0	1	81	80	99	99	94	0	0	0	0	0	0
MT HOTHAM	1849	0	0	0	3	90	89	99	99	97	0	0	0	0	0	0
GELANTIPY	760	0	0	0	0	0	0	0	29	2	1	0	0	0	0	0
COMBIENBAR	640	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0
DUNNS HILL	561	0	0	0	0	0	0	0	42	2	0	0	0	0	0	0
KILMORE GAP	528	0	0	0	0	0	0	0	34	1	0	0	0	0	0	0
EILDON	638	0	0	0	0	0	0	0	49	5	0	0	0	0	0	0

Field Product in GFE/Vis Wx

Elevation corresponding to 50% conditional probability of rain/snow.



For text product: <http://web.bom.gov.au/nob/nmoc/products/dmo/snowdmo/>

Dave Williams Snow Probabilities

- Based on comparison of 9am data (1961 – 2000) where rain or snow was reported at elevated stations with nearby upper air sounding data (within 200km of the stations).

NSW, ACT	⇒	Wagga Wagga Airport
VIC	⇒	Laverton (no longer operational)
TAS	⇒	Hobart Airport

- Logistic regression approach based on these parameters:

Station Elevation

850 hPa Temperature

1000 – 500 hPa Thickness

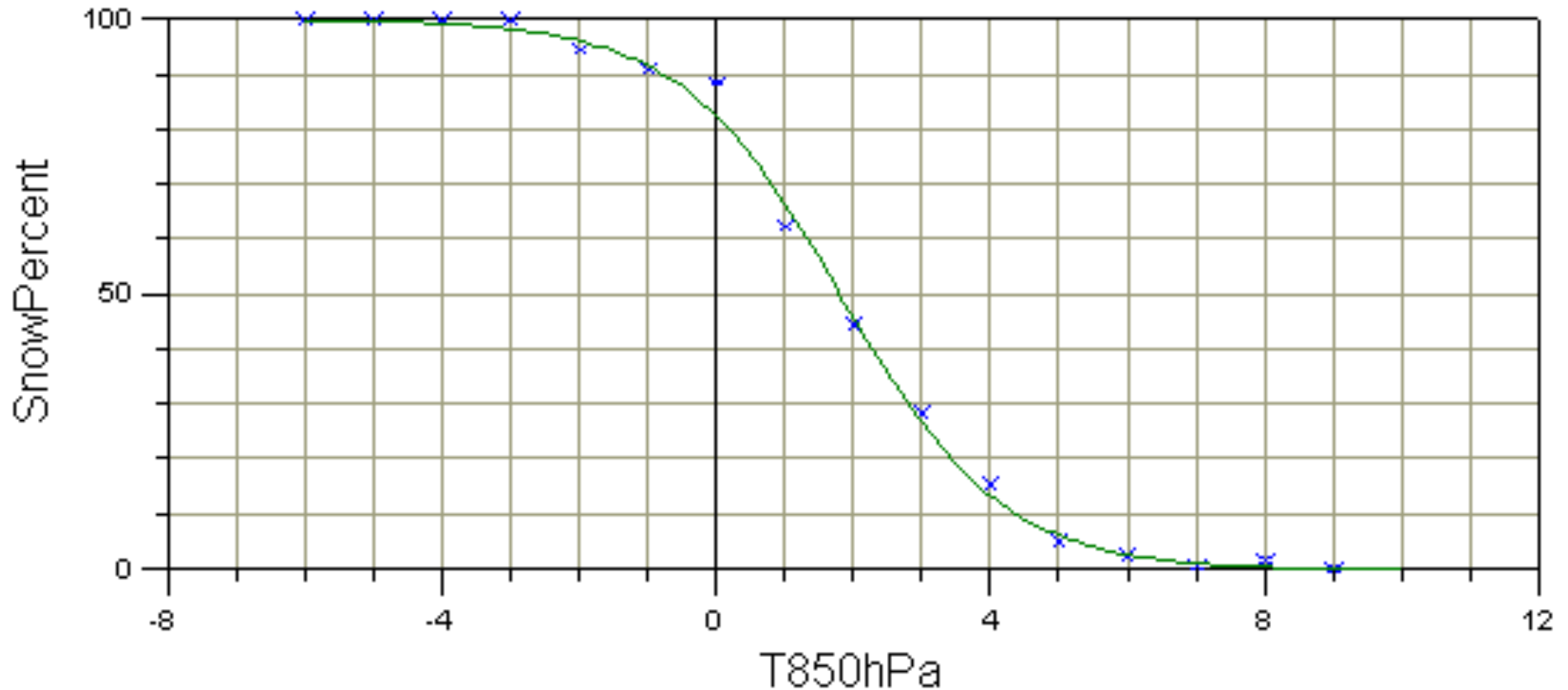
850 hPa Height

Note: On days when $p_{\text{SFC}} < 1000$ hPa the (negative) 1000 hPa height was approximated based on SFC p and T data. The error is less than 5 metres.

Stations used in the Analysis

Location	State	Station ID	Elevation [metres]	Period of data
Cabramurra	NSW	072091	1475	Jan '71 to Mar '99
Charlotte Pass	NSW	071003	1755	Oct '93 to Nov '00
Cradle Valley	TAS	096005	920	Feb '72 to May '77
Honeysuckle Creek	ACT	070241	1116	Jan '71 to Nov '81
Lake St Clair	TAS	096015	735	Feb '61 to Apr '89
Liawenee	TAS	096065	1060	Oct '85 to Dec '00
Maydena	TAS	095011	270	Jan '65 to Jul '92
Mount Barrow	TAS	091131	1323	Jan '65 to Sep '74
Mount Buffalo	VIC	083073	1372	Dec '74 to Mar '80
Mount Hotham	VIC	083081	1750	Dec '77 to Jul '90
Mount Wellington	TAS	094087	1261	Mar '61 to May '72
Orroral Valley	ACT	070206	925	Jan '71 to Aug '85
Perisher Valley	NSW	071072	1735	Jan '86 to Dec '00
Shannon	TAS	096021	939	Mar '64 to Feb '85
Thredbo Crackenback	NSW	071032	1957	Jun '86 to Oct '00
Thredbo Village	NSW	071041	1380	Jan '86 to Dec '00
Tumbarumba	NSW	072043	645	Feb '71 to Nov '00

Cabramurra snow %'s versus Wagga Wagga 850hPa temperature



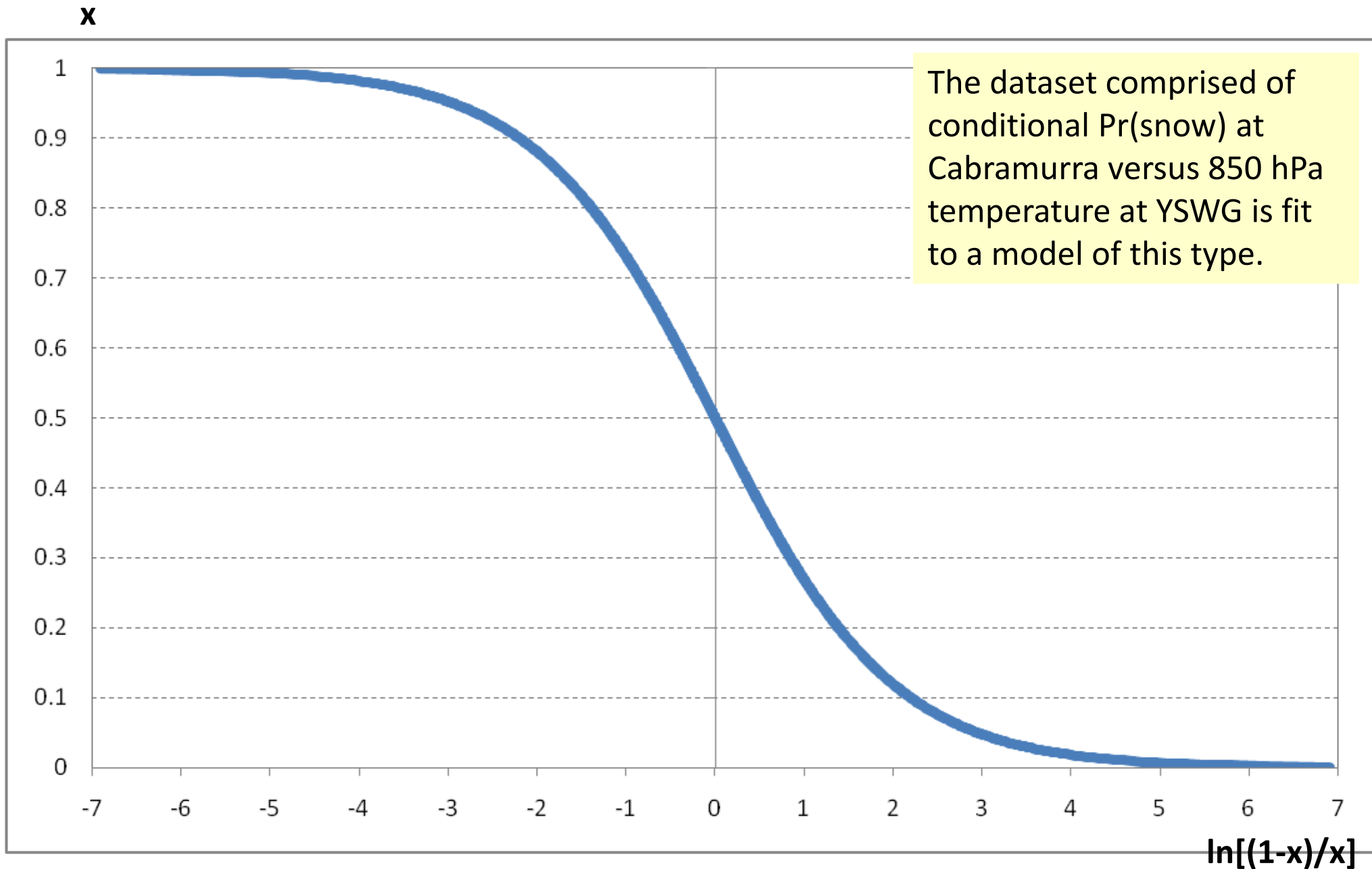
- Cabramurra is SW of Wagga Wagga and SE of Canberra.
- It has an elevation of 1475 metres, which is very close to the 850hPa level for the ICAO Standard Atmosphere (848 hPa).
- These curves were obtained for many different stations.

Map of Cabramurra and Wagga Wagga



Google maps (2016)

Graph of x versus $\ln[(1-x)/x]$



Binary Logistic Model

Let $p = \text{Pr}(\text{Snow})$ $0 < p < 1$

Binary variable p on $[0,1]$ is mapped to $[-\infty, \infty]$ so that regression can be performed on it.

$\Rightarrow q = 1-p = \text{Pr}(\text{Rain})$ $0 < q < 1$

$\Rightarrow q/p = (1-p)/p$ $0 < q/p < \infty$

$\Rightarrow \ln(q/p) = \ln[(1-p)/p]$ $-\infty < \ln(q/p) < \infty$

Form 1 variable binary logistic model as follows:

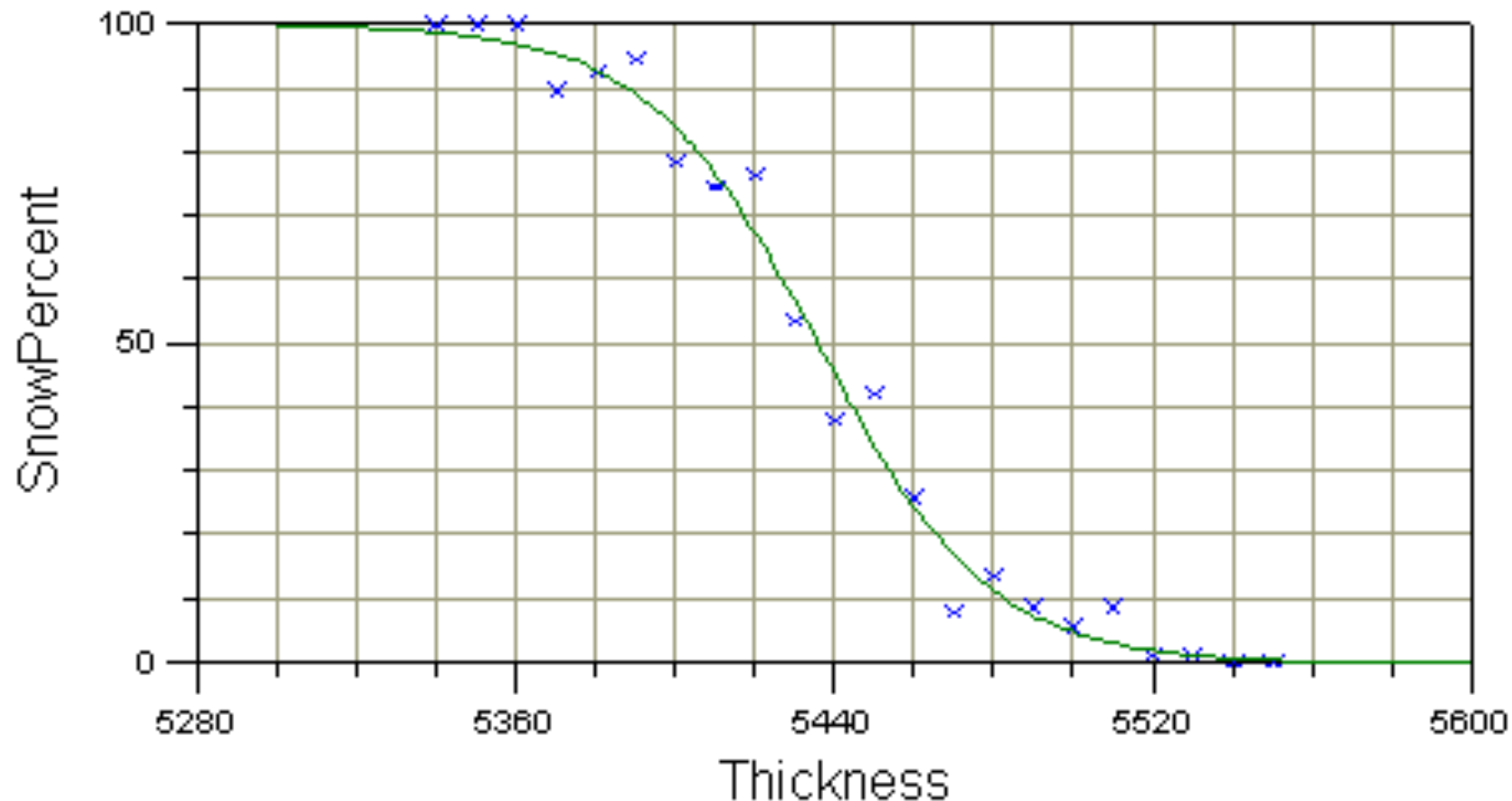
$\Rightarrow \ln[(1-p)/p] = b_0 + b_1x$

$\Rightarrow p = 1 / [1 + \exp(b_0 + b_1x)]$

The coefficients b_0 and b_1 are found by linear regression of the rain/snow data.

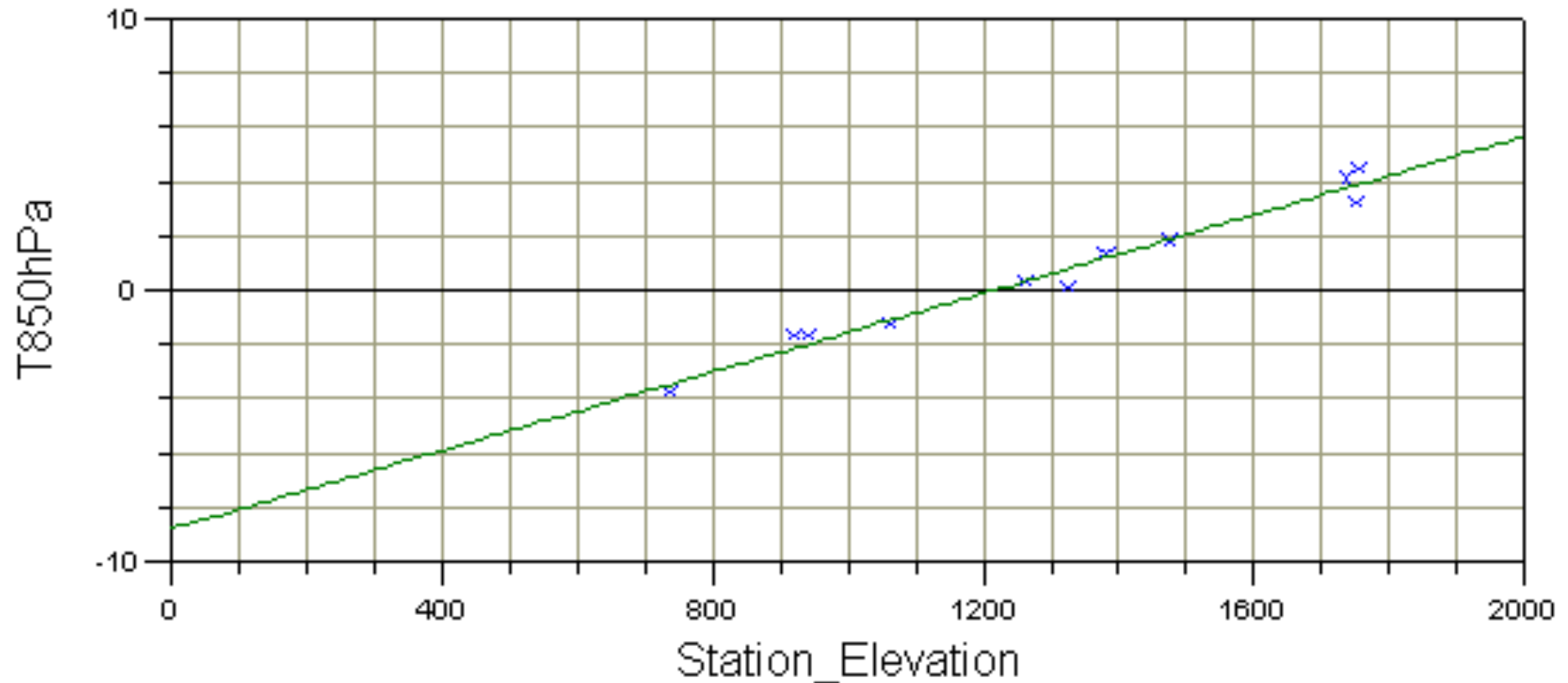
Pr(Snow | Precip) vs $\Delta z(1000-500 \text{ hPa})$

Cabramurra snow %'s versus Wagga Wagga 1000-500hPa thickness



- The relationship between snow probability and 1000-500 hPa thickness at Cabramurra is also very strong, but not quite as strong as the relationship with 850 hPa temperature.

50/50% snow 850hPa temperature versus station elevation.



- Stations that are higher than the 850 hPa level can have $T < 0^{\circ}\text{C}$ when the 850 hPa temperature nearby is $> 0^{\circ}\text{C}$. Hence the 50/50 conditional probability of snow with $T_{850} > 0^{\circ}\text{C}$.
- Lower elevation stations require colder conditions at 850 hPa T to have a 50/50 conditional probability of snow.

Multi-Variable Regression

Full 4-variable regression model has the form:

$$\ln[(1-p)/p] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

Transposing to find the condition probability of snow:

$$\Rightarrow p = 1 / [1 + \exp(b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4)]$$

Where:

- x_1 = Station Elevation
- x_2 = 850 hPa Temperature at YSWG (T_{850})
- x_3 = 850 hPa Height at YSWG (h_{850})
- x_4 = 1000 – 500 hPa Thickness at YSWG (Δz)

50% Snow Levels

- By setting $p=x\%$, this equation can be transformed to determine the elevation at which there is an $x\%$ conditional probability of snow (if precipitation occurs).
- Setting $p=50\%$ gives the median “snow level”.
- $$\text{Snow Level} = - [b_0 + b_2 T_{850} + b_3 h_{850} + b_4 \Delta z] / b_1$$
- Using data from all events at all stations can produce 50% snow level fields from NWP data:
 - 850 Temperature
 - 850 Geopotential Height
 - 1000-500 Thickness.

References

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